Entrega T6

Métodos Matemáticos Para La Física

Problema 1 (4 puntos)

Supongamos que $f(t)$ se define en $(-\pi, \pi]$ como

a) $f(t) = \sin(\pi t)$.
b) $f(t) = \sin^2(t)$.
c) t^3
d) $f(t) = \begin{cases} -1 & \text{if } -\pi < t \leq 0, \\ 1 & \text{if } \ 0 < t \leq \pi. \end{cases}$

Extienda periódicamente y calcule la serie de Fourier.

a) $f(t) = \sin(\pi t) \rightarrow$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} \sin(\pi t)\,dt = \frac{1}{\pi}\left(\left[-\cos(\pi t)\right]_0^{\pi} + \left[-\cos(\pi t)\right]_{-\pi}^{0} \right) = 0$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} \sin(\pi t)\cos(nt) = 0$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} \sin(\pi t)\sin(nt) = \begin{cases} 0 & \text{si } n \neq \pi \\ 1 & \text{si } n = \pi \end{cases}$$

$$f(t) = \sin(\pi t)$$

b) $f(t) = \sin^2(t)$

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} \sin^2(t) = \frac{1}{2\pi}\int_{-\pi}^{\pi} 1 - \cos(2x) = \frac{1}{2\pi}\left[x - \tfrac{1}{2}\sin(2x)\right]_{-\pi}^{\pi} = \frac{1}{2\pi}\left[\pi - -\pi\right] = 1$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} \sin^2(t)\cos(nt)\,dt = \frac{1}{\pi}\int_{-\pi}^{\pi} \frac{1-\cos(2t)}{2}\cos(nt) = \cancelto{0}{\frac{1}{\pi}\int_{-\pi}^{\pi}\frac{1}{2}\cos(nt)} - \frac{1}{\pi}\int_{-\pi}^{\pi}\frac{1}{2}\cos(2t)\cos(nt) = \begin{cases} 0 & \text{si } n \neq 2 \\ -\frac{1}{2} & \text{si } n = 2 \end{cases}$$

Sabiendo que $n = 2$:

$$b_2 = \int_{-\pi}^{\pi} \sin^2(t)\sin(2t) = 0$$

$$f(t) = \frac{a_0}{2} + a_2\cos(2t) + b_2\sin(2t)$$

$$f(t) = \tfrac{1}{2} - \tfrac{1}{2}\cos(2t)$$

c) t^3

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} t^3\,dt = \frac{1}{\pi}\left[\frac{t^4}{4}\right]_{-\pi}^{\pi} = \frac{1}{\pi}\left[\frac{\pi^4}{4} - \frac{\pi^4}{4}\right] = 0$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} t^3\cos(nt)\,dt$$

$$\int t^3\cos(nt)\,dt = \frac{t^3}{n}\,\text{sen}(nt) - \int 3t^2\,\frac{\text{sen}(nt)}{n} = \frac{t^3}{n}\,\text{sen}(nt) + \frac{3t^2}{n^2}\cos(nt) - \int \frac{6t}{n^2}\cos(nt)$$

$u = t^3$, $v = \frac{1}{n}\,\text{sen}(nt)$, $du = 3t^2$, $dv = \cos(nt)$; $u = 3t^2$, $v = -\frac{\cos(nt)}{n^2}$, $du = 6t$, $dv = \frac{\text{sen}(nt)}{n}$; $u = 6t$, $v = -\frac{\text{sen}(nt)}{n^3}$, $du = 6$, $dv = -\frac{\cos(nt)}{n^2}$

$$\int t^3\cos(nt) = \frac{t^3}{n}\,\text{sen}(nt) + \frac{3t^2}{n^2}\cos(nt) + \frac{6t}{n^3}\,\text{sen}(nt) - \int -\frac{6}{n^3}\,\text{sen}(nt)$$

$$\int t^3\cos(nt) = \frac{t^3}{n}\,\text{sen}(nt) + \frac{3t^2}{n^2}\cos(nt) + \frac{6t}{n^3}\,\text{sen}(nt) - \frac{6}{n^4}\cos(nt)$$

$$a_n = \frac{1}{\pi}\left[\cancel{\frac{t^3}{n}\,\text{sen}(nt)}^{\,0} + \frac{3t^2}{n^2}\cos(nt) + \cancel{\frac{6t}{n^3}\,\text{sen}(nt)}^{\,0} - \frac{6}{n^4}\cos(nt)\right]_{-\pi}^{\pi} = \frac{1}{\pi}\left[\frac{3\pi^2}{n^2}\cos(\pi n) - \frac{6}{n^4}\cos(\pi n) - \frac{3\pi^2}{n^2}\cos(\pi n) + \frac{6}{n^4}\cos(\pi n)\right]$$

$$a_n = 0$$

$$b_n = \frac{1}{\pi}\int t^3\,\text{sen}(nt)$$

$$\int t^3\,\text{sen}(nt) = -\frac{t^3}{n}\cos(nt) + \frac{3t^2}{n^2}\,\text{sen}(nt) - \frac{6t}{n^3}\cos(nt) - \frac{6}{n^4}\,\text{sen}(nt)$$

$$b_n = \frac{1}{\pi}\left[-\frac{t^3}{n}\cos(nt) - \frac{6t}{n^3}\cos(nt)\right]_{-\pi}^{\pi} \to \frac{1}{\pi}\left[-\frac{\pi^3}{n}\cos(\pi n) - \frac{6\pi}{n^3}\cos(\pi n) - \frac{\pi^3}{n}\cos(\pi n) - \frac{6\pi}{n^3}\right]$$

$$b_n = \left(-\frac{2\pi^2}{n} - \frac{12}{n^3}\right)(-1)^n = \begin{cases} \frac{2\pi^2}{n} + \frac{12}{n^3} & \text{si } n \text{ es impar} \\ -\frac{2\pi^2}{n} - \frac{12}{n^3} & \text{si } n \text{ es par} \end{cases}$$

$$f(t) = \sum_{n=1}^{\infty}\left(-\frac{2\pi^2}{n} - \frac{12}{n^3}\right)(-1)^n\,\text{sen}(nt) = \sum_{n=par}^{\infty}\left(-\frac{2\pi^2}{n} - \frac{12}{n^3}\right)\text{sen}(nt) + \sum_{n=impar}^{\infty}\left(\frac{2\pi^2}{n} + \frac{12}{n^3}\right)\text{sen}(nt)$$

d) $f(t) = \begin{cases} -1 & \text{si } -\pi < t \leq 0 \\ 1 & \text{si } 0 < t \leq \pi \end{cases}$

$$a_0 = \int_{-\pi}^{\pi} f(t)\,dt = \frac{1}{\pi}\int_0^{\pi} dt + \frac{1}{\pi}\int_{-\pi}^{0} -dt = \frac{1}{\pi}\left[x\right]_0^{\pi} + \frac{1}{\pi}\left[-x\right]_{-\pi}^{0} = \frac{1}{\pi}(\pi) + \frac{1}{\pi}(-\pi) = 0$$

$$a_n = \frac{1}{\pi}\int_0^{\pi} \cos(nt)\,dt + \frac{1}{\pi}\int_{-\pi}^{0} -\cos(nt)\,dt = \frac{1}{\pi}\left[\frac{1}{n}\operatorname{sen}(nt)\right]_0^{\pi} + \frac{1}{\pi}\left[-\frac{1}{n}\operatorname{sen}(nt)\right]_{\pi}^{0} = 0$$

$$b_n = \frac{1}{\pi}\int_0^{\pi} \operatorname{sen}(nt)\,dt + \frac{1}{\pi}\int_{-\pi}^{0} -\operatorname{sen}(nt)\,dt = \frac{1}{\pi}\left(\left[-\frac{1}{n}\cos(nt)\right]_0^{\pi} + \left[\frac{1}{n}\cos(nt)\right]_{-\pi}^{0}\right) = \frac{1}{\pi}\left(-\frac{1}{n}\cos(\pi n) + \frac{1}{n} - \frac{1}{n}\cos(\pi n) + \frac{1}{n}\right)$$

$$b_n = \frac{2}{\pi n}\left(1 - \cos(\pi n)\right) = \frac{2}{\pi n}\left(1 - (-1)^n\right) \longrightarrow b_n = \begin{cases} 0 & \text{si } n \text{ es par} \\ \frac{4}{\pi n} & \text{si } n \text{ es impar} \end{cases}$$

$$f(t) = \sum_{n=\text{impar}}^{\infty} \frac{4}{n\pi} \operatorname{sen}(n\pi)$$

Made with Goodnotes

Problema 2 (2 puntos)

Sea $f(t) = t^2$ para $-2 < t \leq 2$ extendida periódicamente.

a) Calcule la serie de Fourier para $f(t)$..

b) Escriba la serie explícitamente hasta el tercer armónico.

$$L = 2$$

$$a)\ a_0 = \frac{1}{L}\int_{-L}^{L} f(t)\,dt = \frac{1}{2}\int_{-2}^{2} t^2\,dt = \frac{1}{2}\left[\frac{t^3}{3}\right]_{-2}^{2} = \frac{1}{2}\cdot\frac{8}{3} - \frac{1}{2}\cdot -\frac{8}{3} = \frac{8}{3}$$

$$a_n = \frac{1}{L}\int_{-L}^{L} f(t)\cos\left(\frac{n\pi t}{L}\right)dt = \frac{1}{2}\int_{-2}^{2} t^2\cos\left(\frac{n\pi}{2}t\right)$$

$$\int t^2\cos\left(\frac{n\pi}{2}t\right) = \frac{2t^2}{n\pi}\operatorname{sen}\left(\frac{n\pi}{2}t\right) + \frac{8t}{n^2\pi^2}\cos\left(\frac{n\pi}{2}t\right) - \int\frac{8}{n^2\pi^2}\cos\left(\frac{n\pi}{2}t\right)dt = \frac{2t^2}{n\pi}\operatorname{sen}\left(\frac{n\pi}{2}t\right) + \frac{8t}{n^2\pi^2}\cos\left(\frac{n\pi}{t}\right) + \frac{16}{n^3\pi^3}\operatorname{sen}\left(\frac{n\pi}{2}t\right)$$

$$a_n = \frac{1}{2}\left[\frac{2t^2}{n\pi}\operatorname{sen}\left(\frac{n\pi}{2}t\right) + \frac{8t}{n^2\pi^2}\cos\left(\frac{n\pi}{2}t\right) + \frac{16}{n^3\pi^3}\operatorname{sen}\left(\frac{n\pi}{2}t\right)\right]_{-2}^{2} = \frac{8}{(n\pi)^2}\cos(n\pi) + \frac{8}{(n\pi)^2}\cos(n\pi)$$

$$a_n = \frac{8}{n^2\pi^2}\left(\cos n\pi + \cos n\pi\right) = \frac{16}{(n\pi)^2}(-1)^n$$

$$b_n = \frac{1}{2}\int_{-2}^{2} t^2\operatorname{sen}\left(\frac{n\pi}{2}t\right) = \left[-\frac{t^2}{n\pi}\cos\left(\frac{n\pi}{2}t\right) - \frac{8}{n^3\pi^3}\cos\left(\frac{n\pi}{2}t\right)\right]_{-2}^{2} = -\frac{4}{n\pi}\cos(n\pi) - \frac{8}{(n\pi)^3}\cos(n\pi) + \frac{4}{n\pi}\cos(n\pi) + \frac{8}{(n\pi)^3}\cos(n\pi)$$

$$\int t^2\operatorname{sen}\left(\frac{n\pi}{2}t\right) = -\frac{2t^2}{n\pi}\cos\left(\frac{n\pi}{2}t\right) + \int\frac{8t}{n^2\pi^2}\operatorname{sen}\left(\frac{n\pi}{2}t\right)dt - \frac{16}{n^3\pi^3}\cos\left(\frac{n\pi}{2}t\right)$$

$$b_n = 0$$

$$f(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n\cos\left(\frac{n\pi}{L}t\right) + b_n\operatorname{sen}\left(\frac{n\pi}{L}t\right) = \frac{4}{3} + \sum_{n=1}^{\infty}\frac{16}{(n\pi)^2}(-1)^n\cos\left(\frac{n\pi}{2}t\right)$$

b) Armónico 1 $= \frac{4}{3} - \frac{16}{\pi^2}\cos\left(\frac{\pi}{2}t\right)$

Armónico 2 $= \frac{4}{3} - \frac{16}{\pi^2}\cos\left(\frac{\pi}{2}t\right) + \frac{4}{\pi^2}\cos(\pi t)$

Armónico 3 $= \frac{4}{3} - \frac{16}{\pi^2}\cos\left(\frac{\pi}{2}t\right) + \frac{4}{\pi^2}\cos(\pi t) - \frac{16}{9\pi^2}\cos\left(\frac{3\pi}{2}t\right)$

Problema 3 (2 puntos)

Sea $f(t) = t \quad /3$ en el intervalo $0 \leq t < 3$.

a) Encuentra la serie de Fourier de la extensión periódica par.

b) Encuentra la serie de Fourier de la extensión periódica impar.

$L = 3/2$

$$a_0 = 2/3 \int_{-3/2}^{3/2} t/3 \, dt \rightarrow a_0 = 2/3 \left[\frac{t^2}{6} \right]_{-3/2}^{3/2} = 2/3 \left(\frac{(3/2)^2}{6} - \frac{(-3/2)^2}{6} \right) = 0$$

$$a_n = 2/3 \int_{-3/2}^{3/2} t/3 \cos\left(\frac{2n\pi}{3} t\right)$$

$$\int t/3 \cos\left(\frac{2n\pi}{3} t\right) = \frac{1}{2} \cdot \frac{t}{n\pi} \operatorname{sen}\left(\frac{2n\pi}{3} t\right) - \int \frac{1}{2n\pi} \cdot \operatorname{sen}\left(\frac{2n\pi}{3} t\right) dt = \frac{1}{2} \cdot \frac{t}{n\pi} \cdot \operatorname{sen}\left(\frac{2n\pi}{3} t\right) + \frac{3}{4} \cdot \frac{1}{(n\pi)^2} \cdot \cos\left(\frac{2n\pi}{3} t\right)$$

$u = t/3 \qquad v = \frac{3}{2n\pi} \operatorname{sen}\left(\frac{2n\pi}{3} t\right)$

$du = 1/3 \qquad dv = \cos\left(\frac{2n\pi}{3} t\right)$

$$a_n = 2/3 \left[\frac{t}{2n\pi} \operatorname{sen}\left(\frac{2n\pi}{3} t\right) + \frac{3}{4(n^2\pi^2)} \cos\left(\frac{2n\pi}{3} t\right) \right]_{-3/2}^{3/2} = \frac{1}{2n^2\pi^2} \left(\cos(\pi n) - \cos(\pi n) \right) = 0$$

$$b_n = 2/3 \int_{-3/2}^{3/2} t/3 \operatorname{sen}\left(\frac{2n\pi}{3} t\right) dt = \frac{2}{3} \left[\frac{t}{2n\pi} \cos\left(\frac{2n\pi}{3} t\right) \right]_{-3/2}^{3/2} = \frac{1}{2n\pi} \left(\cos(\pi n) + \cos(\pi n) \right) = \frac{1}{n\pi} (-1)^n$$

$$\int t/3 \operatorname{sen}\left(\frac{2n\pi}{3} t\right) dt = -\frac{t}{2n\pi} \cos\left(\frac{2n\pi}{3} t\right) + \frac{3}{4(n^2\pi^2)} \operatorname{sen}\left(\frac{2n\pi}{3} t\right)$$

$$b_n = \begin{cases} -\frac{1}{n\pi} & \text{si } n \text{ impar} \\ \frac{1}{n\pi} & \text{si } n \text{ par} \end{cases}$$

a) Fourier extensión par $= \sum_{n=par}^{\infty} \frac{1}{n\pi} \operatorname{sen}\left(\frac{2n\pi}{3} t\right)$

b) Fourier extensión impar $= \sum_{n=impar}^{\infty} -\frac{1}{n\pi} \operatorname{sen}\left(\frac{2n\pi}{3} t\right)$

Problema 4 (2 puntos)

a) Dada la EDO $x''(t) + 2x(t) = f(t)$, donde $f(t) = t$ definida sobre el intervalo $0 < t < 1$. Dadas las condiciones de contorno $x(1) = 0$, $x(1) = 0$ encontrar la solución particular.

b) Repetir el problema para las condiciones de contorno $x'(0) = 0$, $x'(1) = 0$.

a) $x'' + 2x = t$ donde $0 < t < 1$

$L = 1/2 \rightarrow$ Extensión impar

$$b_n = 2\int_0^1 t \sin(n\omega t) = 2\left[-\frac{t}{n\omega}\cos(n\omega t) + \frac{1}{n^2\omega^2}\operatorname{sen}(n\omega t)\right]_0^1 = -\frac{2}{n\omega}(-1)^n$$

$$f(t) = \sum -\frac{2(-1)^n}{n\omega}\operatorname{sen}(n\omega t)$$

$$x(t) = \sum A_n \operatorname{sen}(n\omega t), \quad x'(t) = \sum (n\omega) A_n \cos(n\omega t), \quad x''(t) = -\sum (n\omega)^2 A_n \operatorname{sen}(n\omega t)$$

$$-\sum A_n (n\omega)^2 \operatorname{sen}(n\omega t) + 2\sum A_n \operatorname{sen}(n\omega t) = \sum -\frac{2(-1)^n}{n\omega}\operatorname{sen}(n\omega t)$$

$$\left(-(n\omega)^2 + 2\right) A_n = -\frac{2(-1)^n}{(n\omega)} \rightarrow A_n = -\frac{2(-1)^n}{-(n\omega)^3 + 2\omega n}$$

$$x(t) = \sum_{n=1}^{\infty} \frac{-2(-1)^n}{-(n\omega^3) + 2\omega n}\operatorname{sen}(n\omega t)$$

b) Como involucra derivadas: Extensión par

$$a_0 = \int_0^1 t\,dt = 1/2$$

$$a_n = 2\int t\cos(n\omega t)\,dt = 2\left[\frac{t}{n\omega}\operatorname{sen}(n\omega t) + \frac{1}{n^2\omega^2}\cos(n\omega t)\right]_0^1 = \frac{2}{n^2\omega^2}\left[\cos(\omega n) - 1\right]$$

$$a_n = \frac{2((-1)^n - 1)}{(n\omega^2)}$$

$x(t) = \sum A_n \cos(n\omega t)$ $\qquad x''(t) = -\sum (n\omega^2) A_n \cos(n\omega t)$

$$-\sum (n\omega)^2 \cos(n\omega t) A_n + 2\sum A_n \cos(n\omega t) = \sum \frac{2((-1)^n - 1)}{(n\omega)^2}$$

$$A_n\left(-(n\omega)^2 + 2\right) = \frac{2((-1)^n - 1)}{(n\omega)^2} \rightarrow A_n = \frac{2((-1)^n - 1)}{-(n\omega)^4 + 2(n\omega)^2}$$